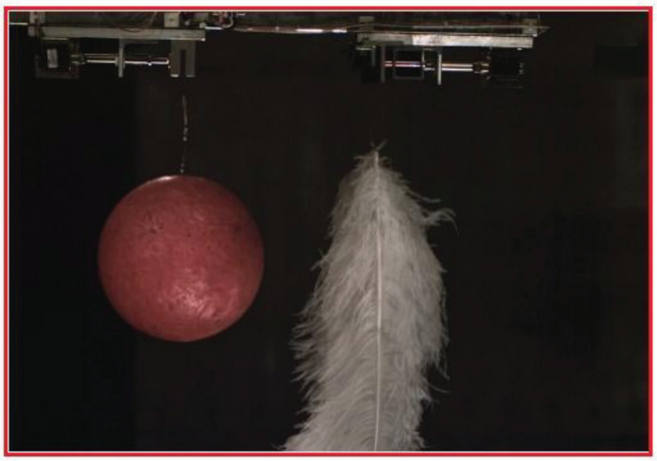
By following equation (2.19) both iron ball and feather reach the Earth with the same speed. It is given by

$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 10}$$
$$= \sqrt{200}\, m\, s^{-1} \approx 14.14\, m\, s^{-1}$$

DML67

Galileo concluded that in vaccum all objects fall with the same acceleration g and reach the ground at the same time.



https://www.youtube.com/watch?v=E43-CfukEgs&t=5s

EXAMPLE 2.35

Is it possible to measure the depth of a well using kinematic equations?



Consider a well without water, of some depth d. Take a small object (for example lemon) and a stopwatch. When you drop the lemon, start the stop watch. As soon as the lemon touches the bottom of the well, stop the watch. Note the time taken by the lemon to reach the bottom and denote the time as t.

Since the initial velocity of lemon $u = 0$ and the acceleration due to gravity g is constant over the well, we can use the equations of motion for constant acceleration.

$$s = ut + \frac{1}{2}at^2$$

Since $u = 0, s = d, a = g$ (Since we choose the y axis downwards), Then

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